AIDEN CHENEY

MECHANICAL ENGINEERING | THE UNIVERSITY OF TEXAS AT AUSTIN

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FSAE REAR WHEEL HUB ASSEMBLY

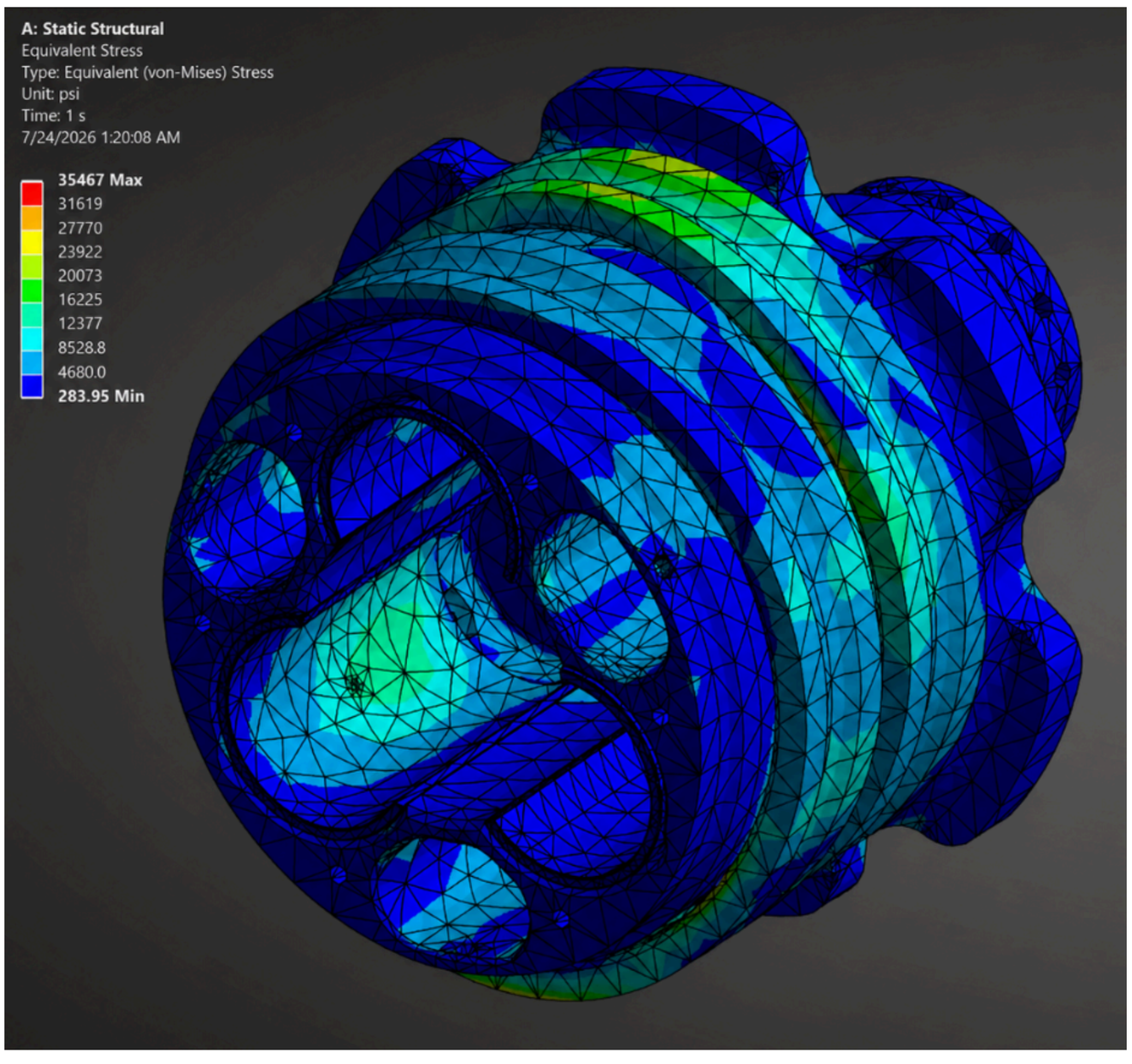
Longhorn Racing Electric | Aug–Dec 2025

Project Scope: Owned the complete development of a lightweight, high-stiffness rear wheel hub assembly from requirements and CAD through FEA, tolerance definition, machining, vehicle integration, and 600+ miles of on-car testing.

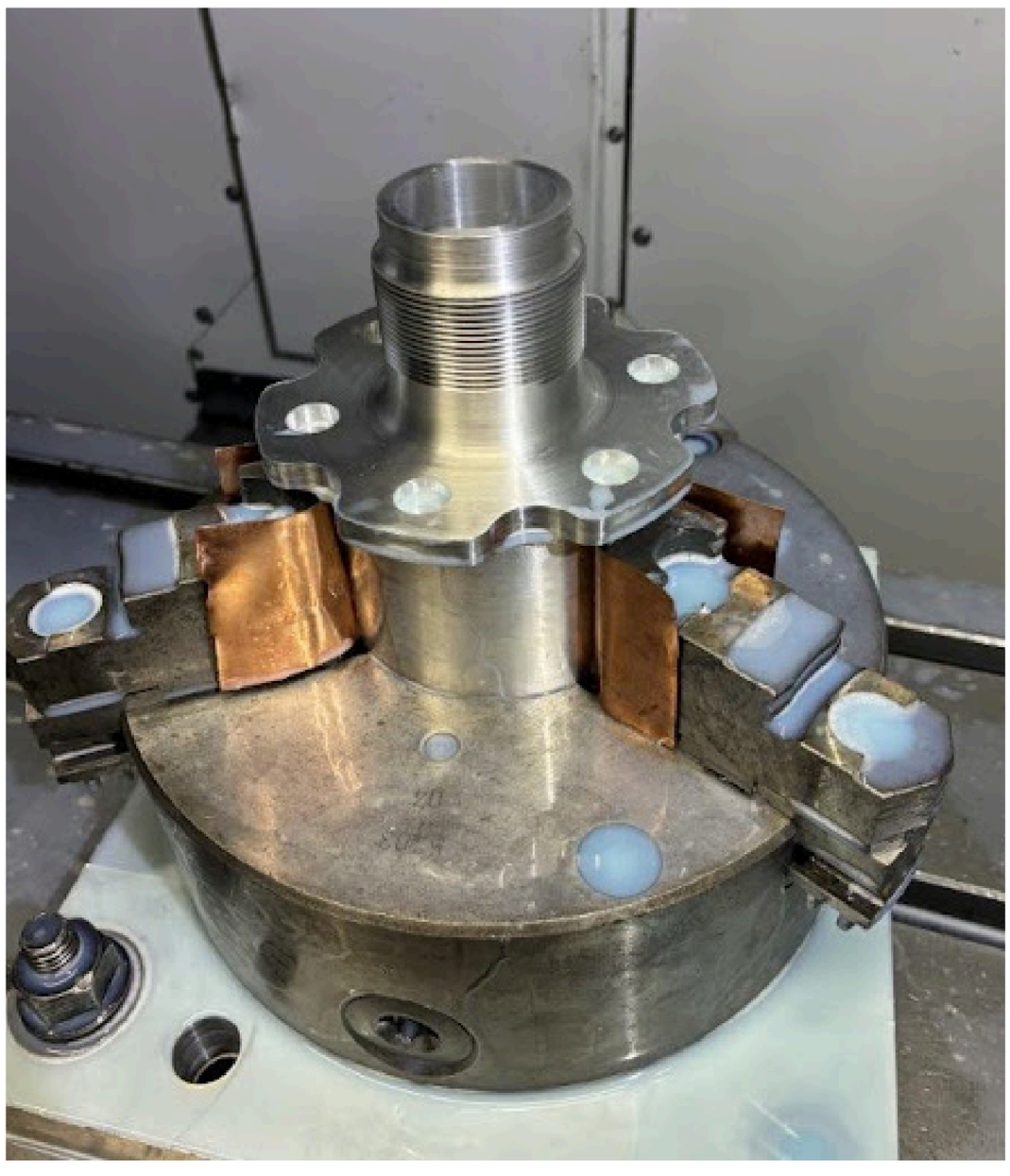
- Redistributed mass from FEA-confirmed low-stress regions (inner bore, hub-wheel interface) into shimmed tripod-bearing contact zone after first-iteration hotspots
- Verified fatigue life via Miner's rule across 4 load cases (braking/cornering/combined/bump), Marin-adjusted S-N data, Goodman mean-stress correction
- Released ASME Y14.5 drawing: functional datums, press-fit reqs, worst-case tolerance stack
- 2 CNC setups, 1 re-fixture, full Fusion 360 CAM toolpath set

Metric	Result
Mass	1.31 → 1.12 lb (–14.5%)
FOS	2.06 (2g/2g/3g + braking torque)
Fatigue life	500,000 cycles (Miner's rule)
Validation	600+ on-car miles, 0 failures

Combined-load FEA | FOS 2.06



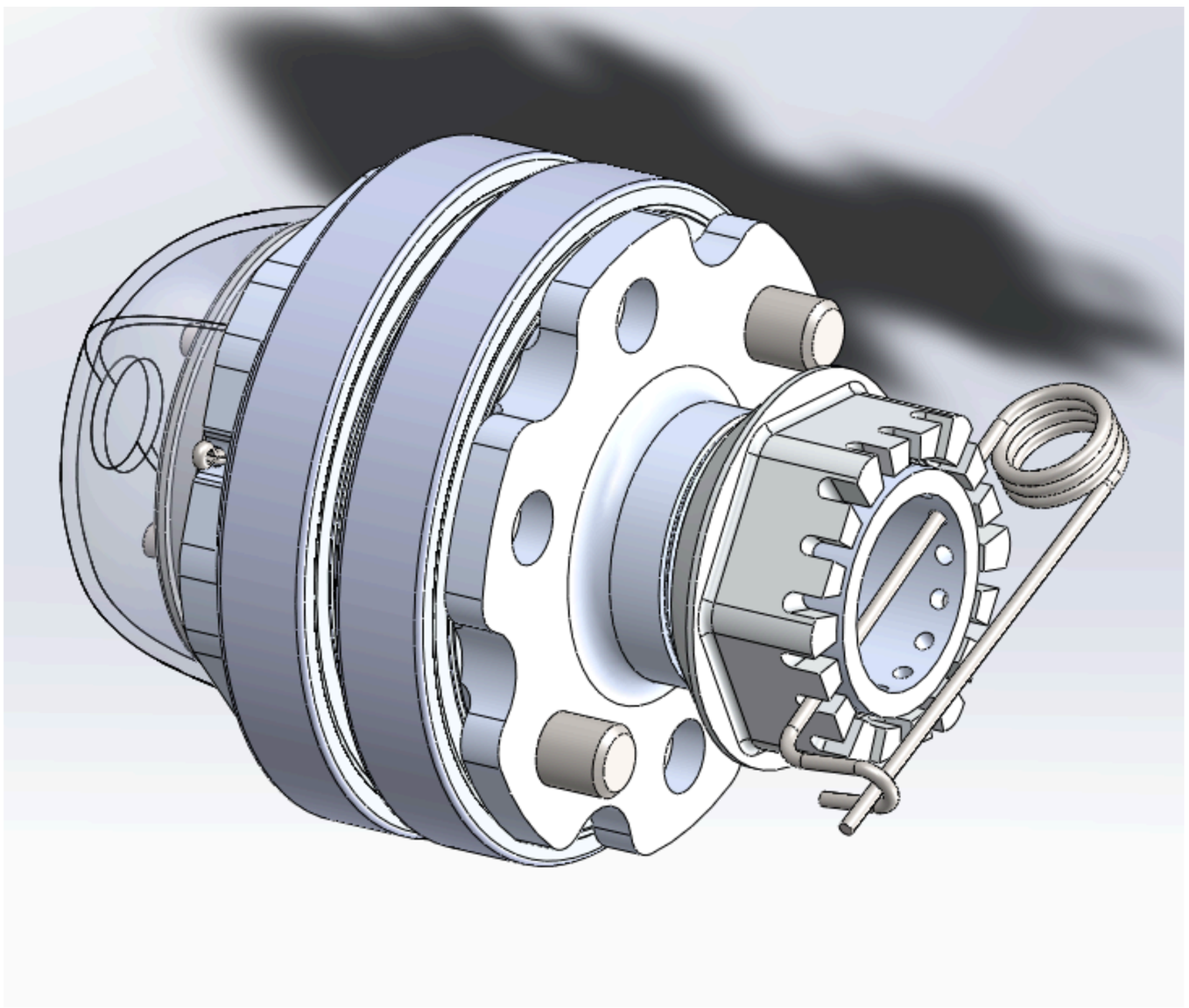
CNC machining — setup 2/2



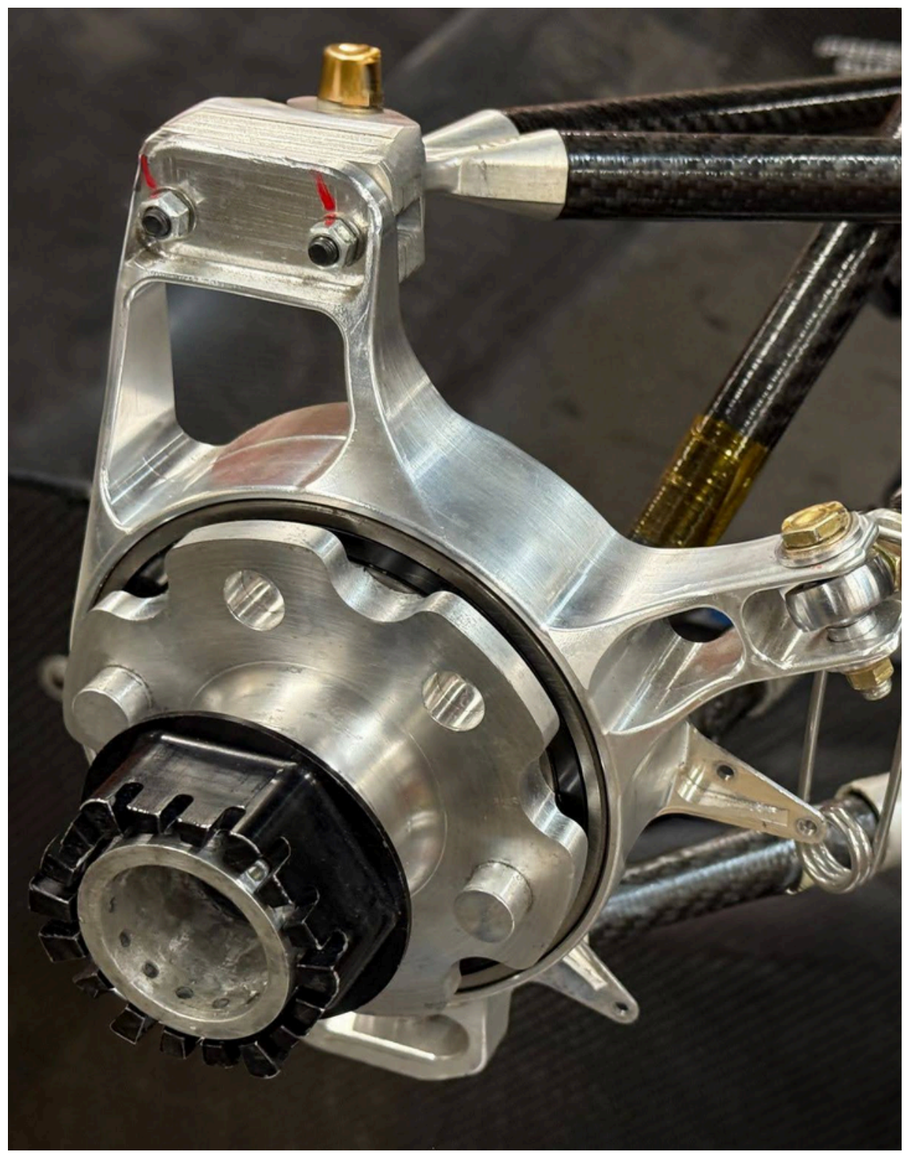
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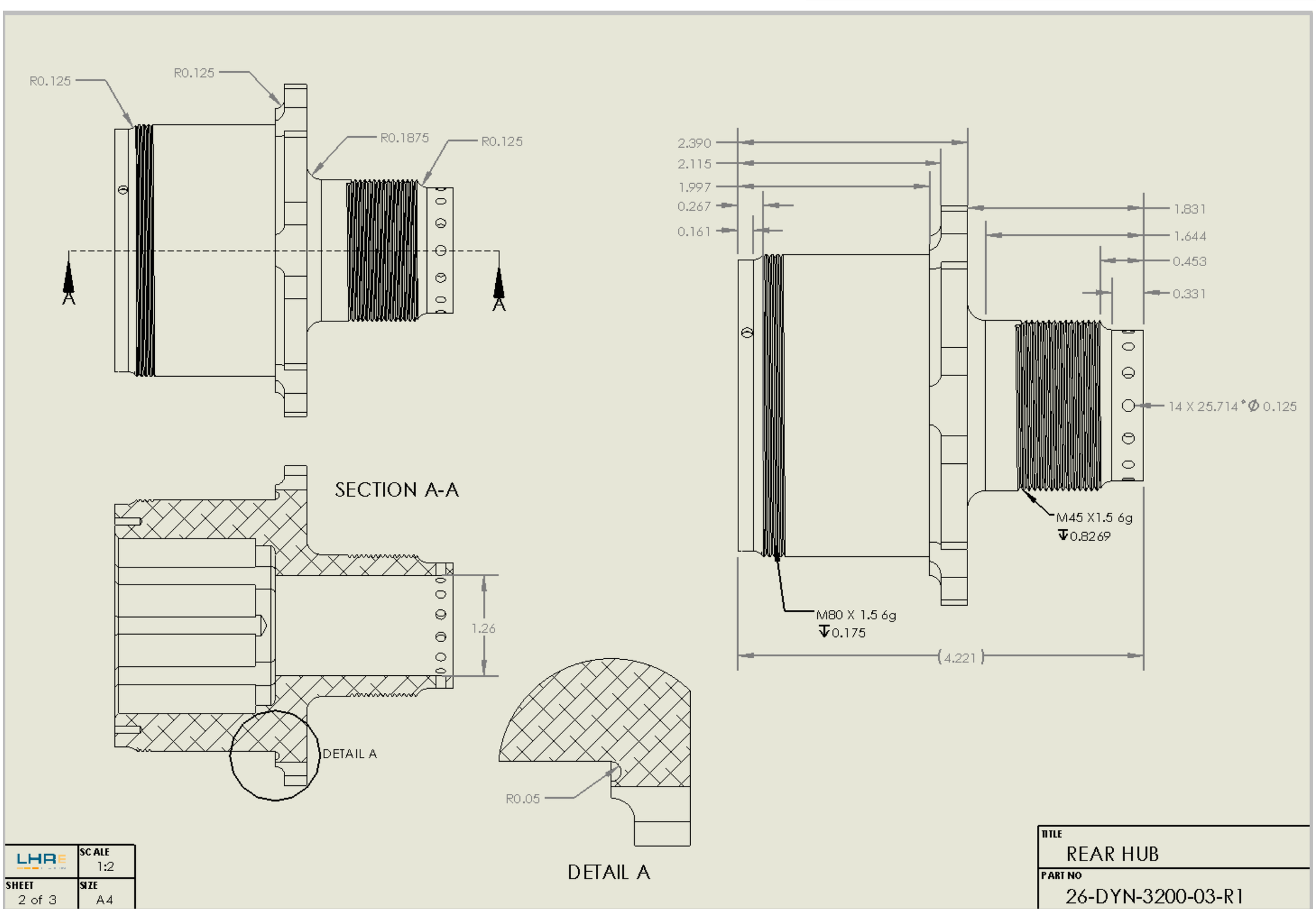
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Hub + bearing/preload assembly



Installed rear hub | 600+ miles





ASME Y14.5 Manufacturing Drawing

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FSAE SUSPENSION DESIGN, SIMULATION & FATIGUE ANALYSIS

Independent Projects | May–Jul 2026

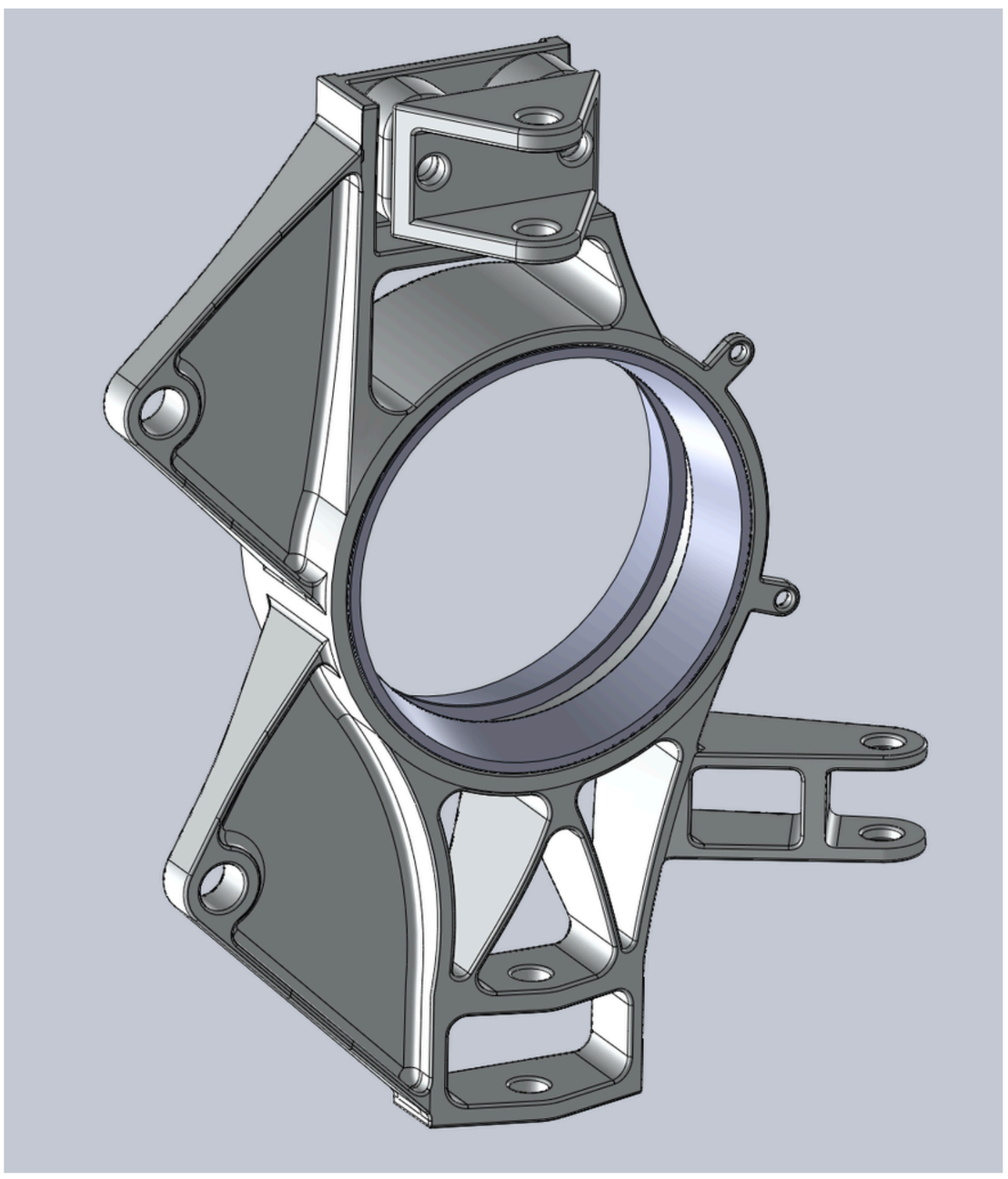
Project Scope: A self-directed project built around the team's real suspension hardpoints, taking on a more complex CAD/FEA problem than my prior work by designing and iterating a billet front upright from scratch. Built a custom data logger to learn the real physics and dynamics of the car firsthand, then applied that real measured data to a rainflow-counting fatigue-life analysis, closing the loop from load to stress life.

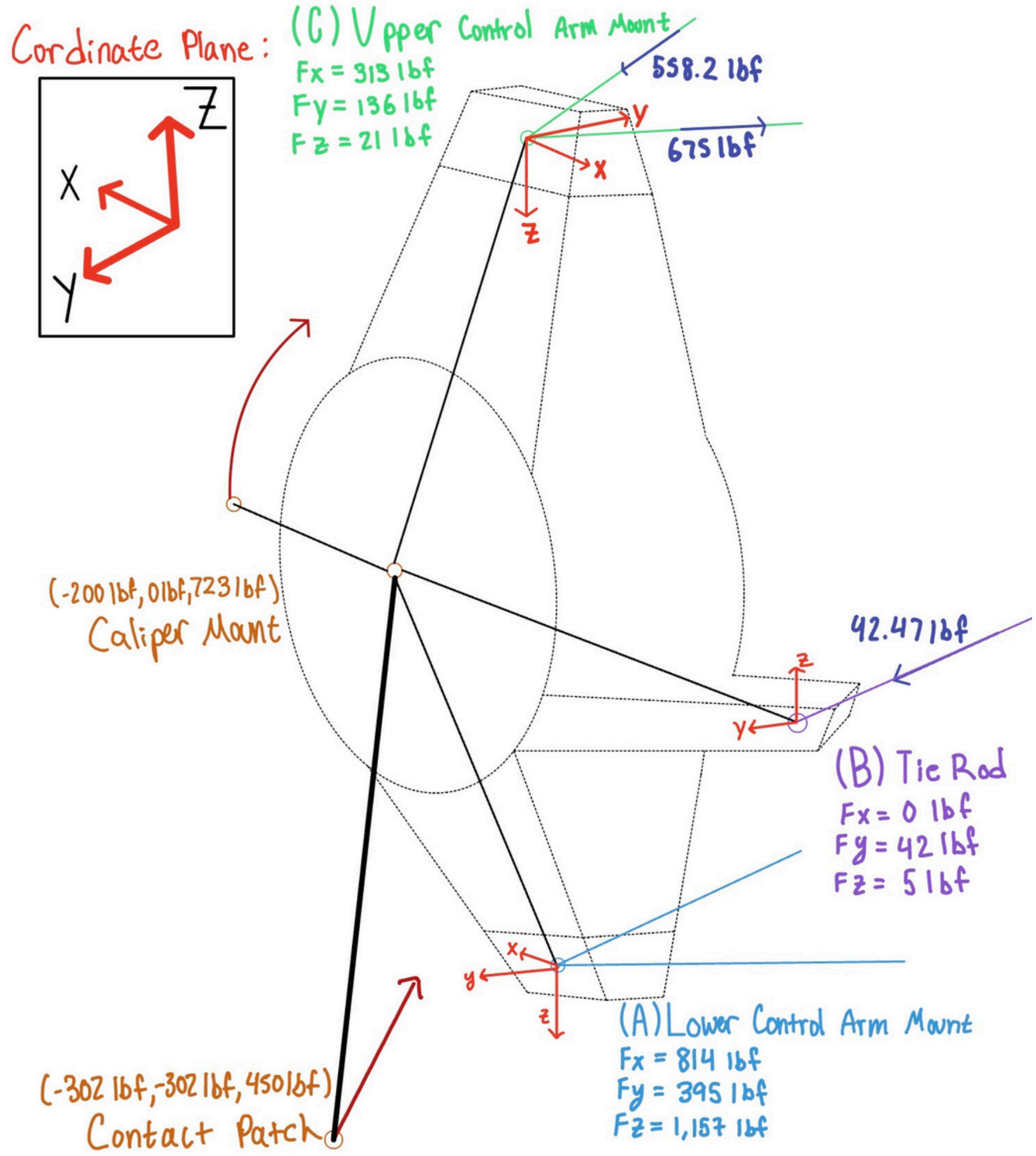
Billet Upright :

- Triangulated-rib geometry from real team hardpoints, tension/compression load paths, not bending bendingDiagnosed iteration-1 singularity as a modeling artifact before trusting the result
- I-beam caliper-mount rib cut deflection 0.098→0.020 in for +0.01 lb only

Iteration	FOS	Max stress	Deflection	Mass
1	—	singularity	0.098 in	0.52 lb
2	1.57	46,496 psi	0.020 in	0.48 lb
3 (final)	1.90	38,387 psi	0.009 in	0.49 lb

Final upright CAD | 0.49 lb

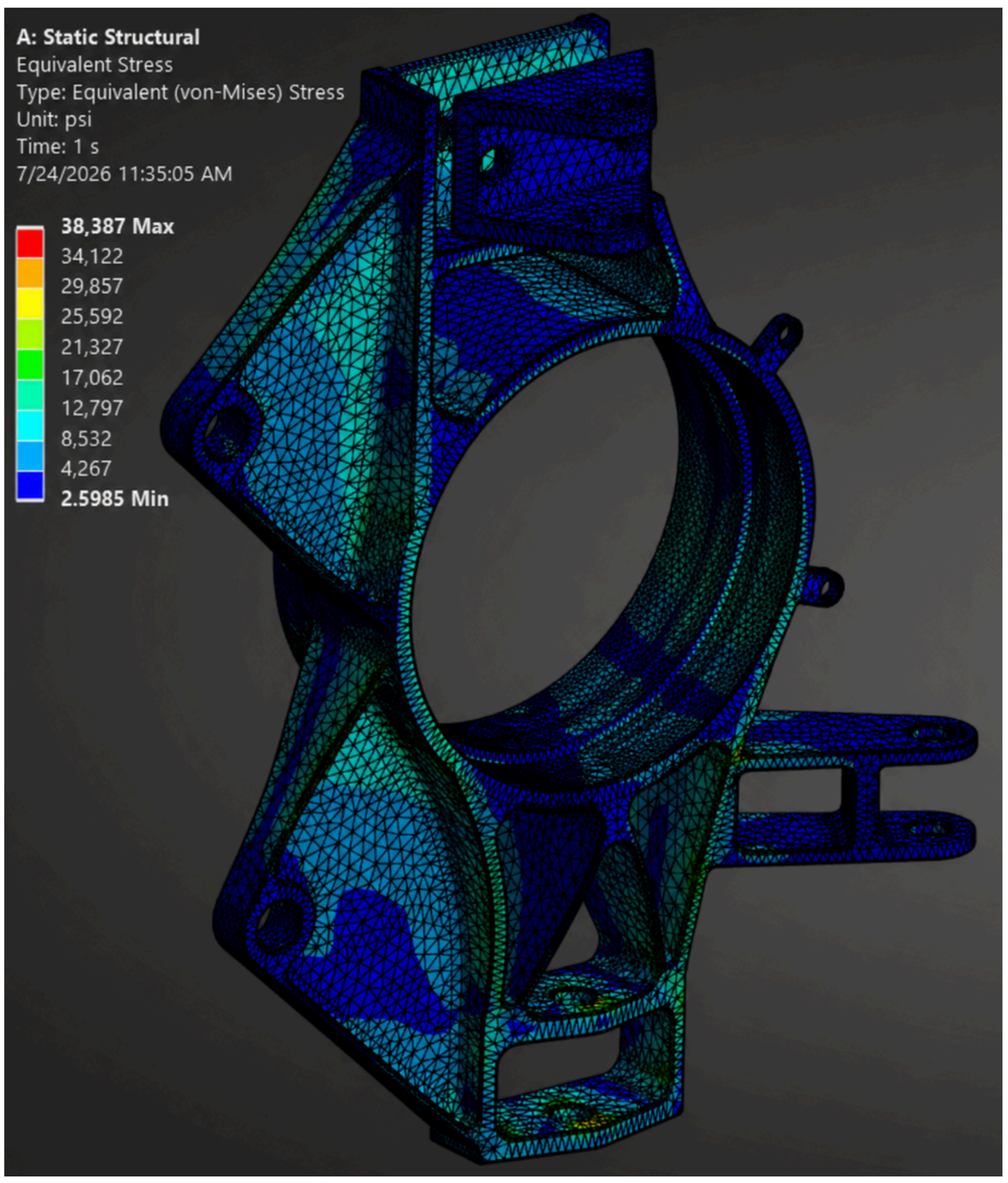




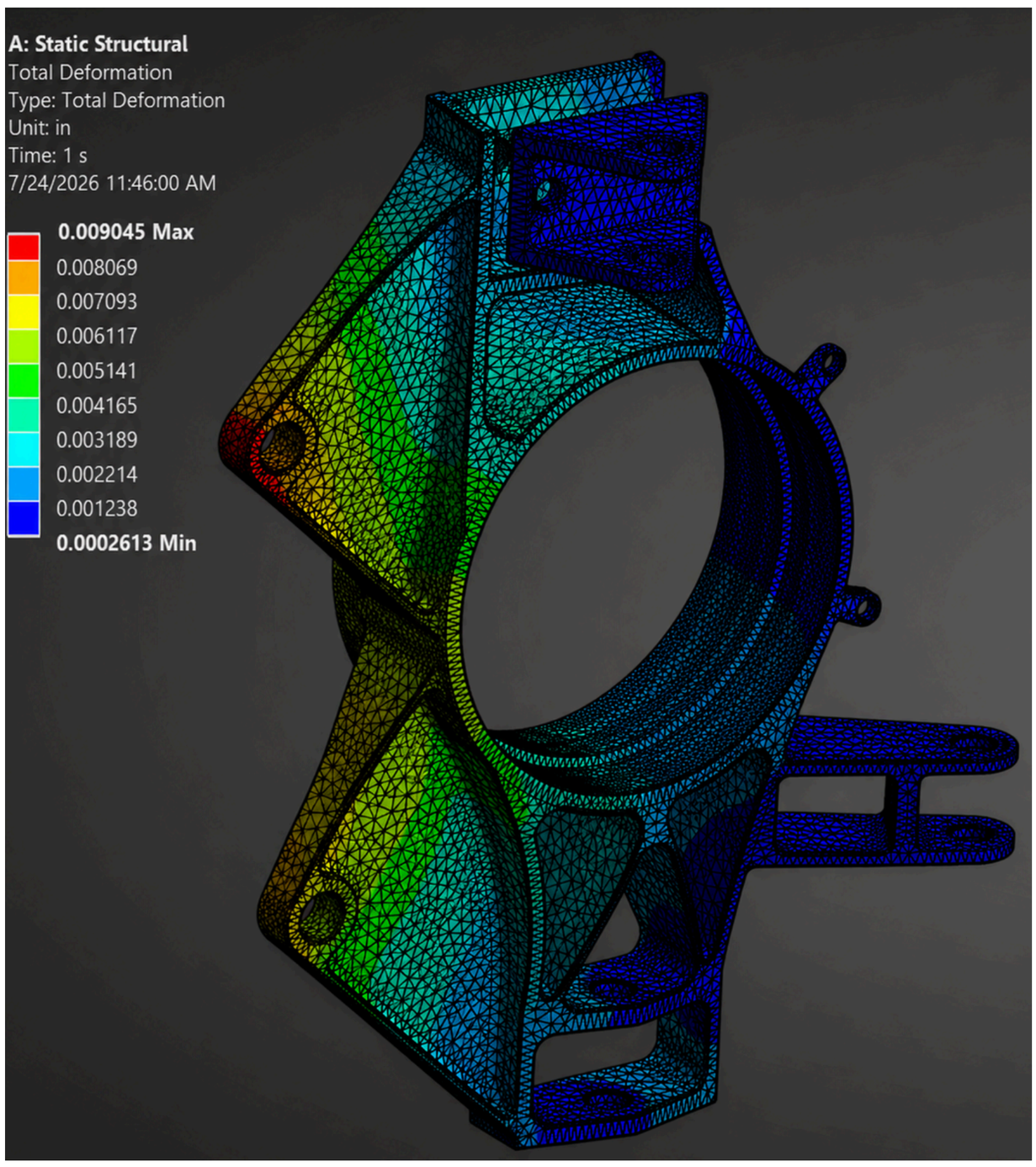
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Equivalent stress | 38.4 ksi max

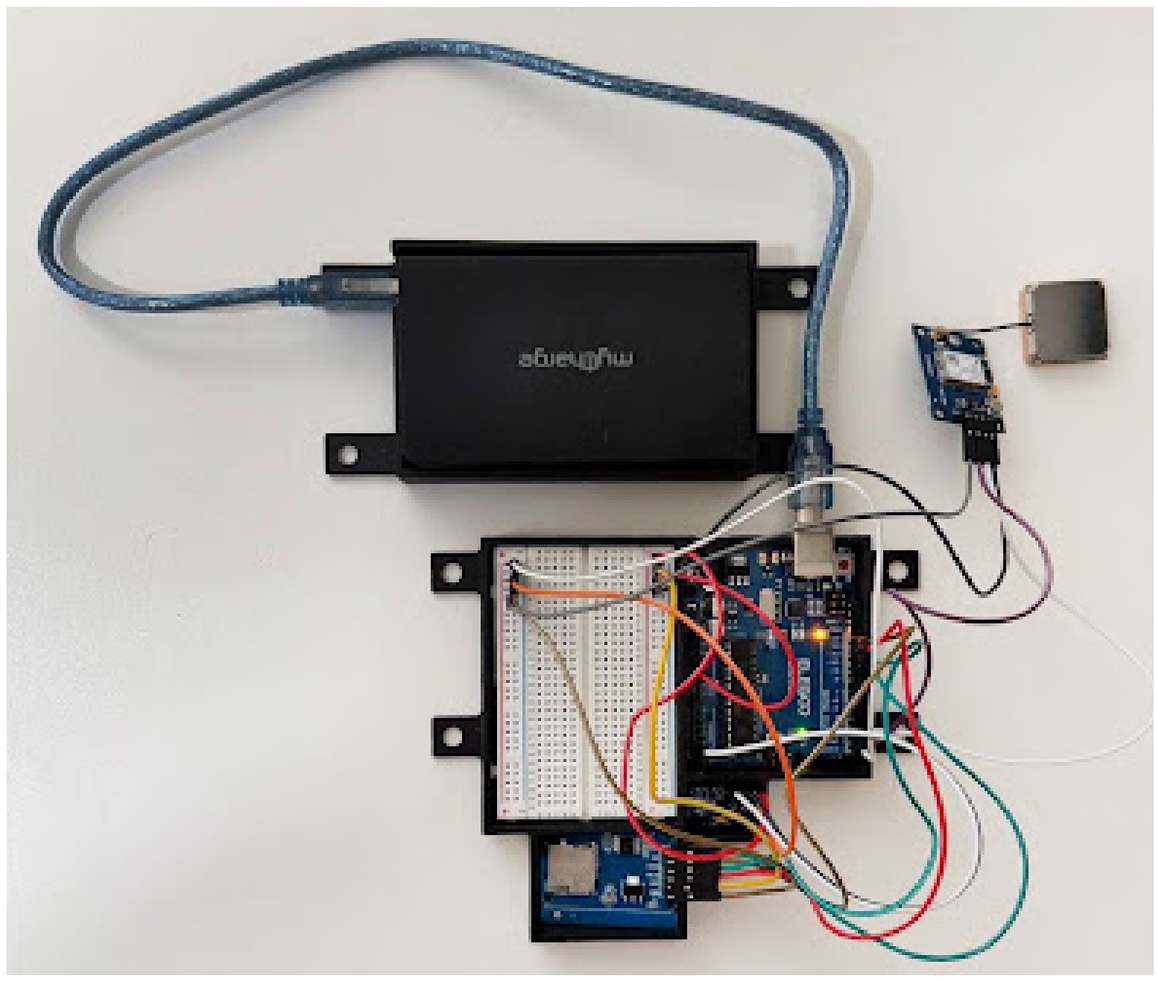


Total deformation | 0.009 in max



Tracksim and Data Logger:

- Python 4-tire load-sensitive grip solver, Pacejka fit to Round 9 TTC data: self-consistent, not fixed-friction (fixed-friction overpredicts high-load-corner grip)
- 10.32s predicted lap (parametric track), a setup-comparison tool
- Validated sensor-to-stress pipeline end-to-end on real logged data; flagged and excluded vibration-artifact spikes before trusting it on hardware



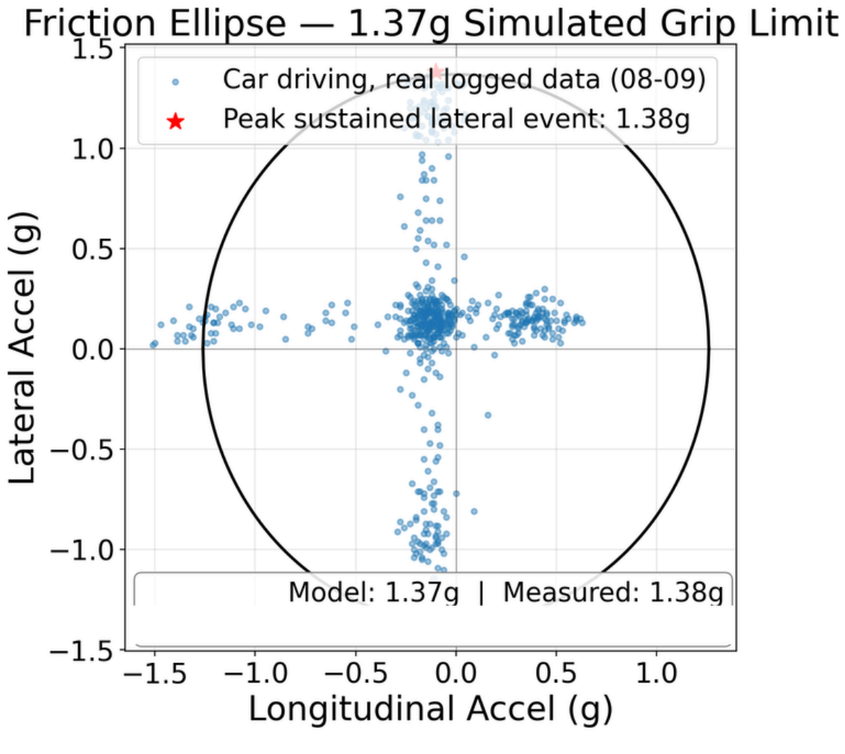
Arduino Data Logger

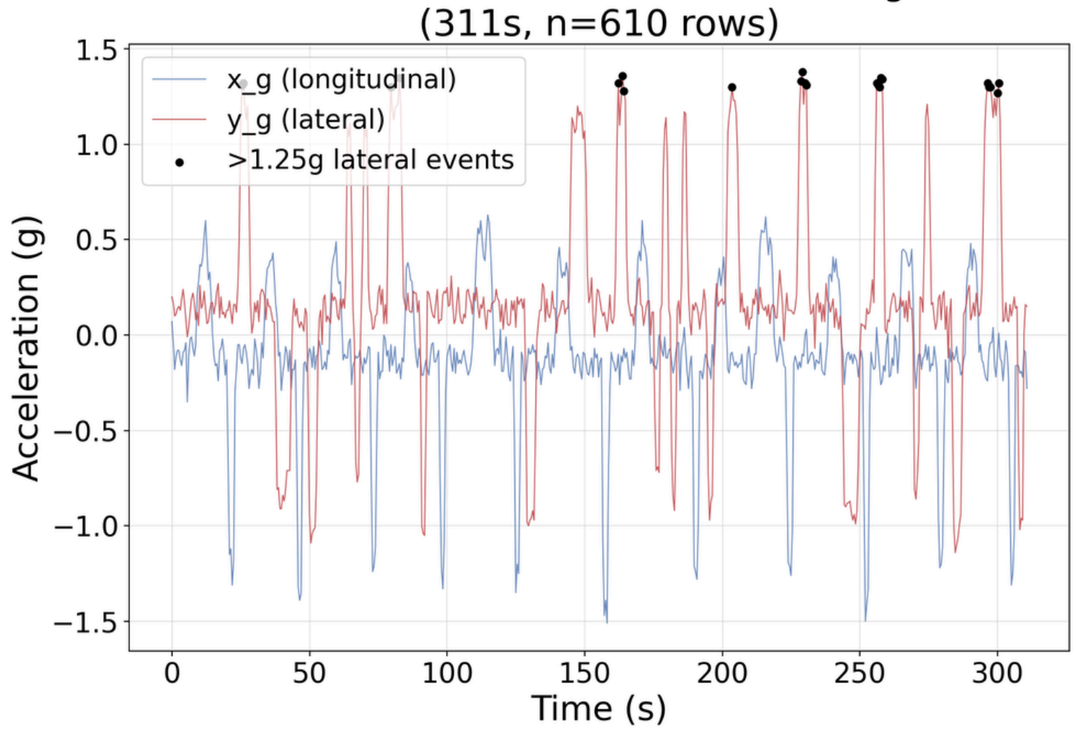
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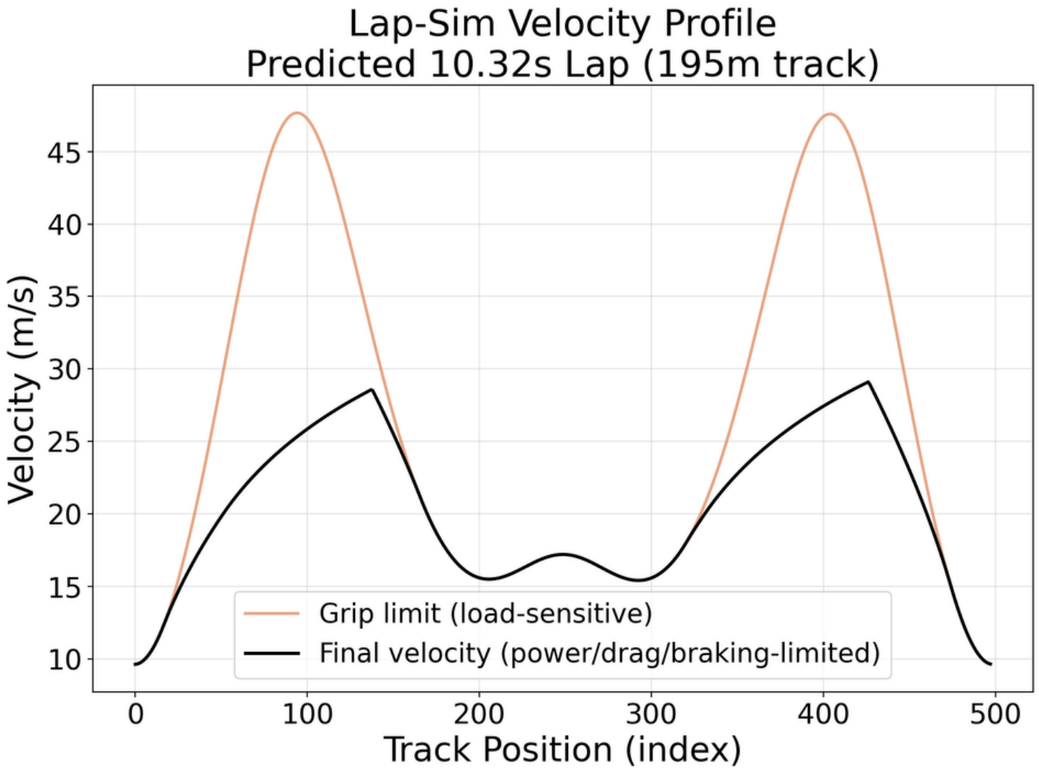
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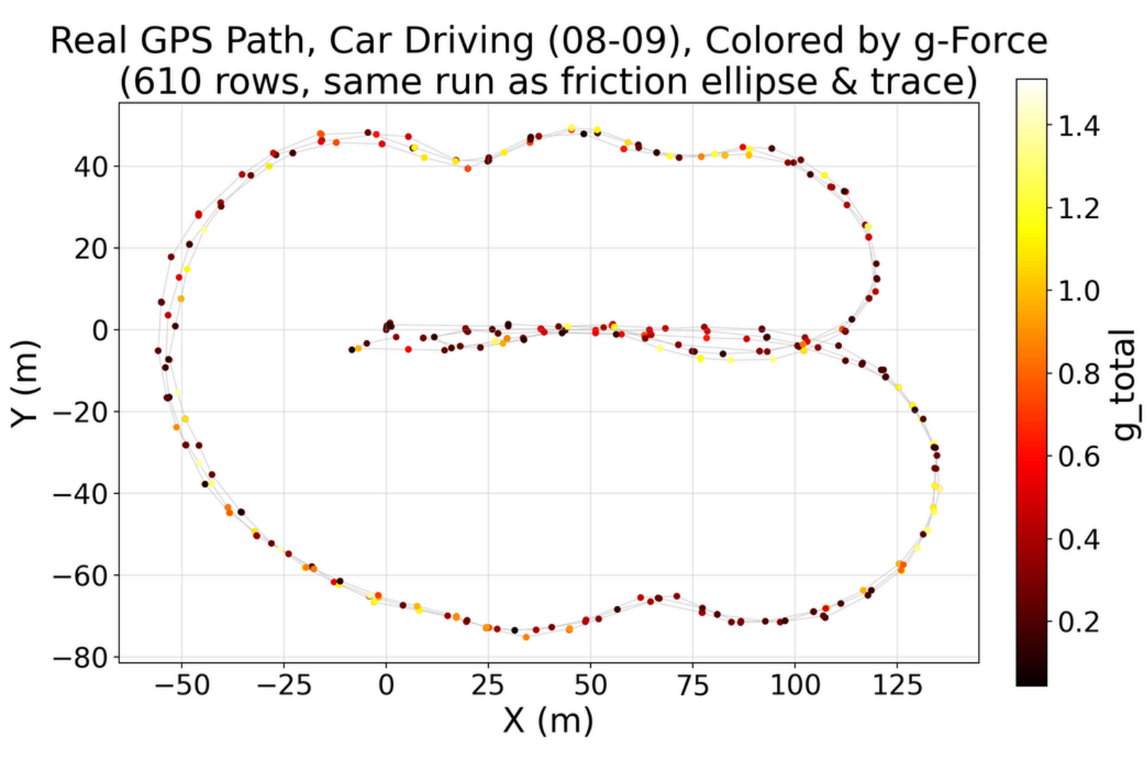
Fatigue Analysis (Rainflow)

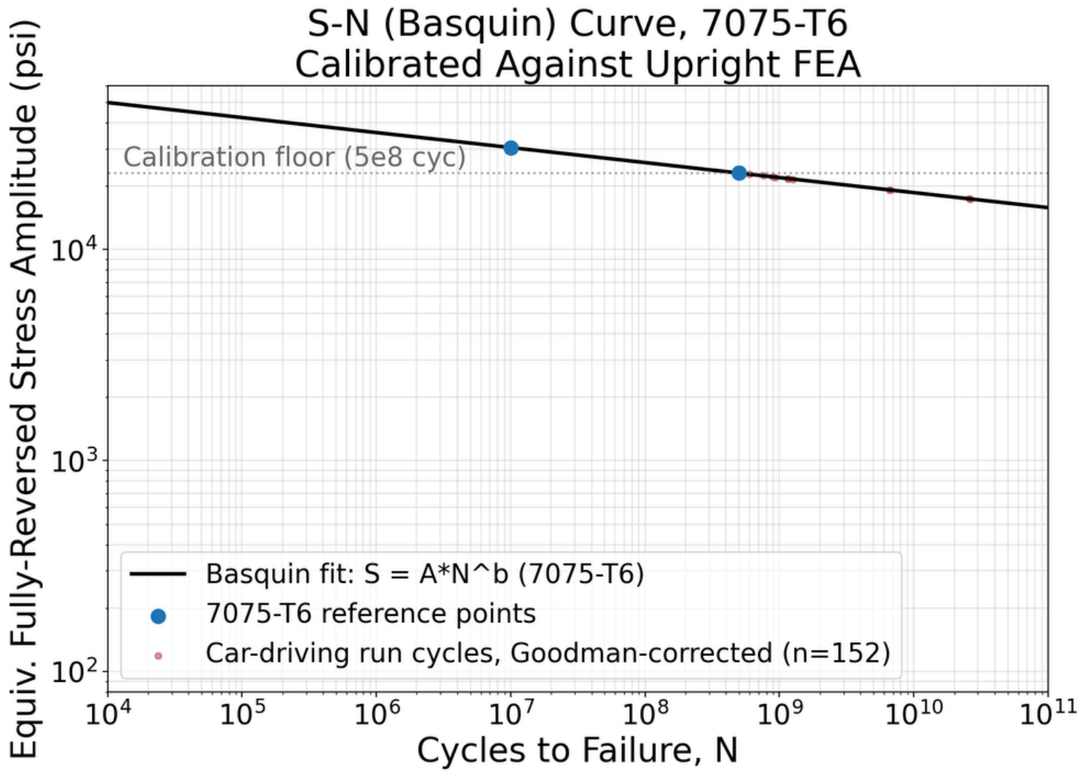
- ASTM E1049 rainflow + Basquin S-N + Goodman correction + Miner's rule, calibrated to FEA stress-per-g
- 5 real driving runs collected; only cleanest run in calibrated range (22,497 psi, within 2.5% of S-N floor), other 4 excluded, not extrapolated

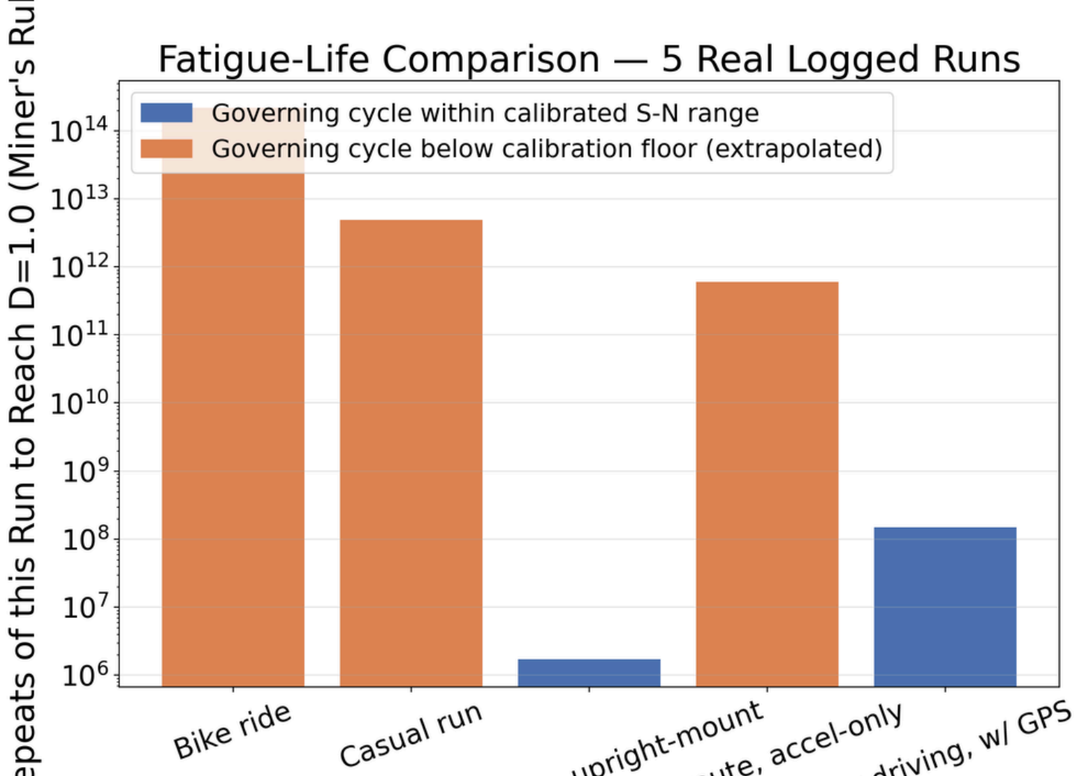












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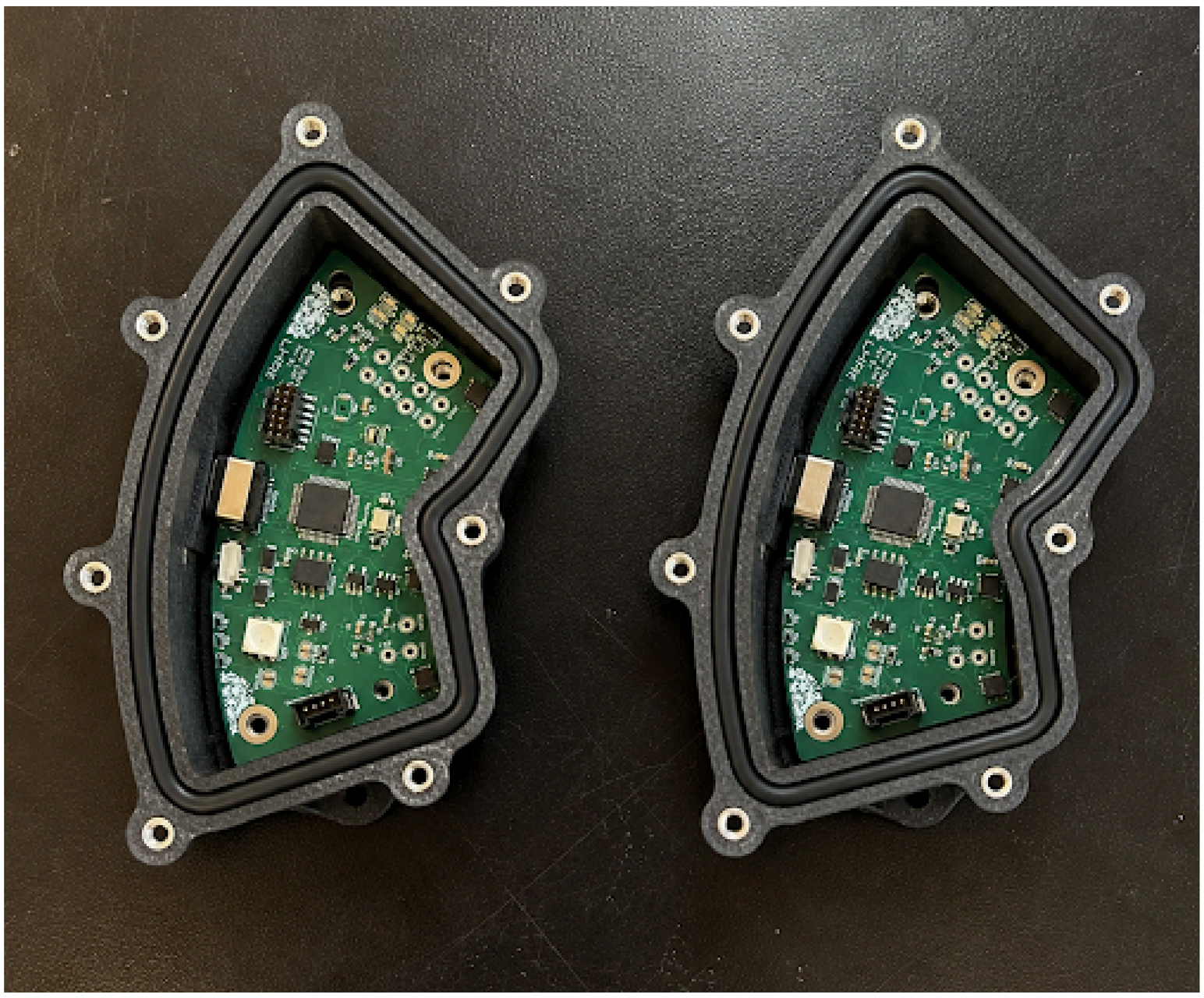
HALL-EFFECT SENSOR ENCLOSURE & MAGNETIC INTEGRATION

Longhorn Racing Electric | Aug 2025–Jan 2026

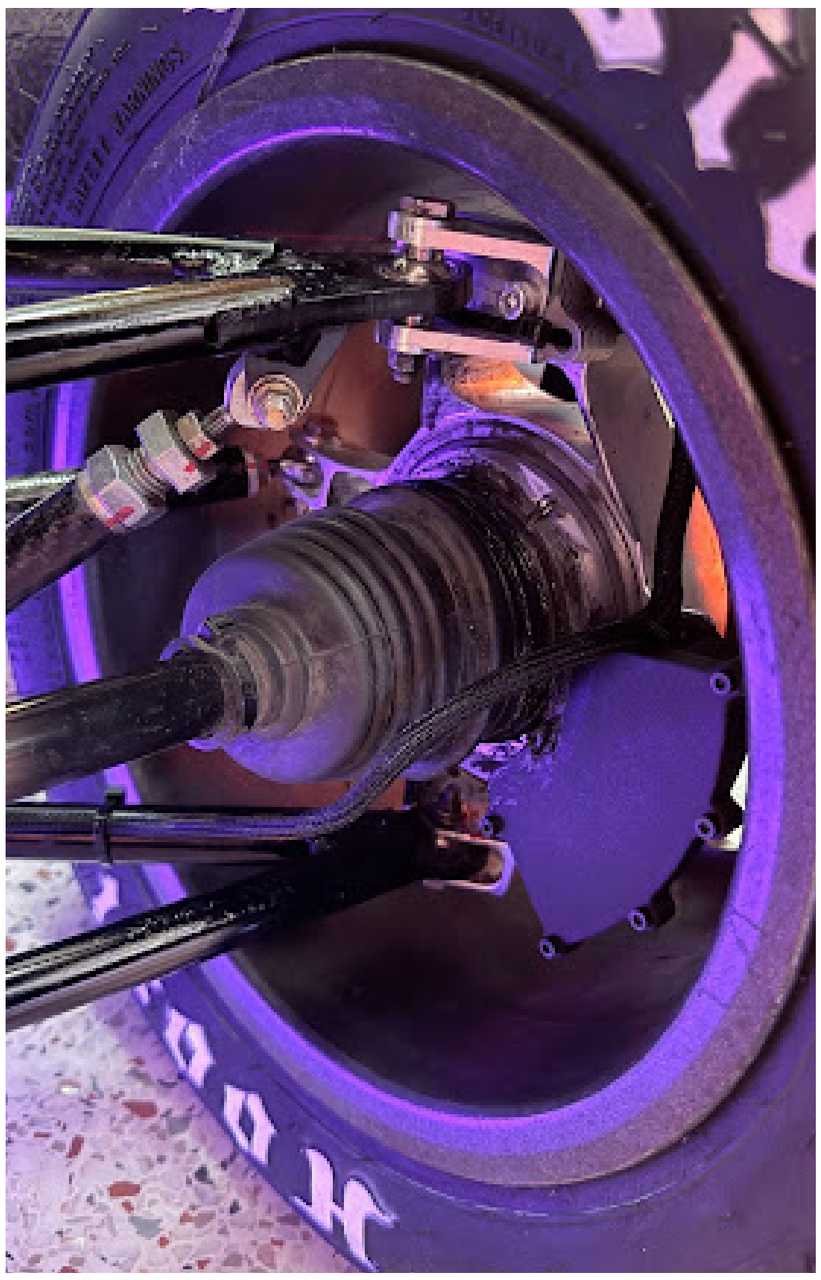
- Sealed PCB enclosure, integrated O-ring grooves, IP67 water-ingress rating in unsprung packaging
- Pivot-and-arcuate-slot PCB mount for fine Hall-effect sensor alignment to rotating magnets
- Magnet strength/placement sized via magnetic flux calc sheet, informing custom-machined magnet-carrying preload nut
- Fastener count and lid thickness tuned alongside O-ring depth; found to affect sealing as much as the O-ring itself

Test	Result
Gland depth trials	1/16 in (failed), 1/8 in, 3/32 in (selected, min footprint)
Immersion test	1 m, 30 min, IP67 criterion, 0 leakage
Signal margin	Magnet strength +20%, reduced angular misalignment

Assembled wheel-speed enclosures



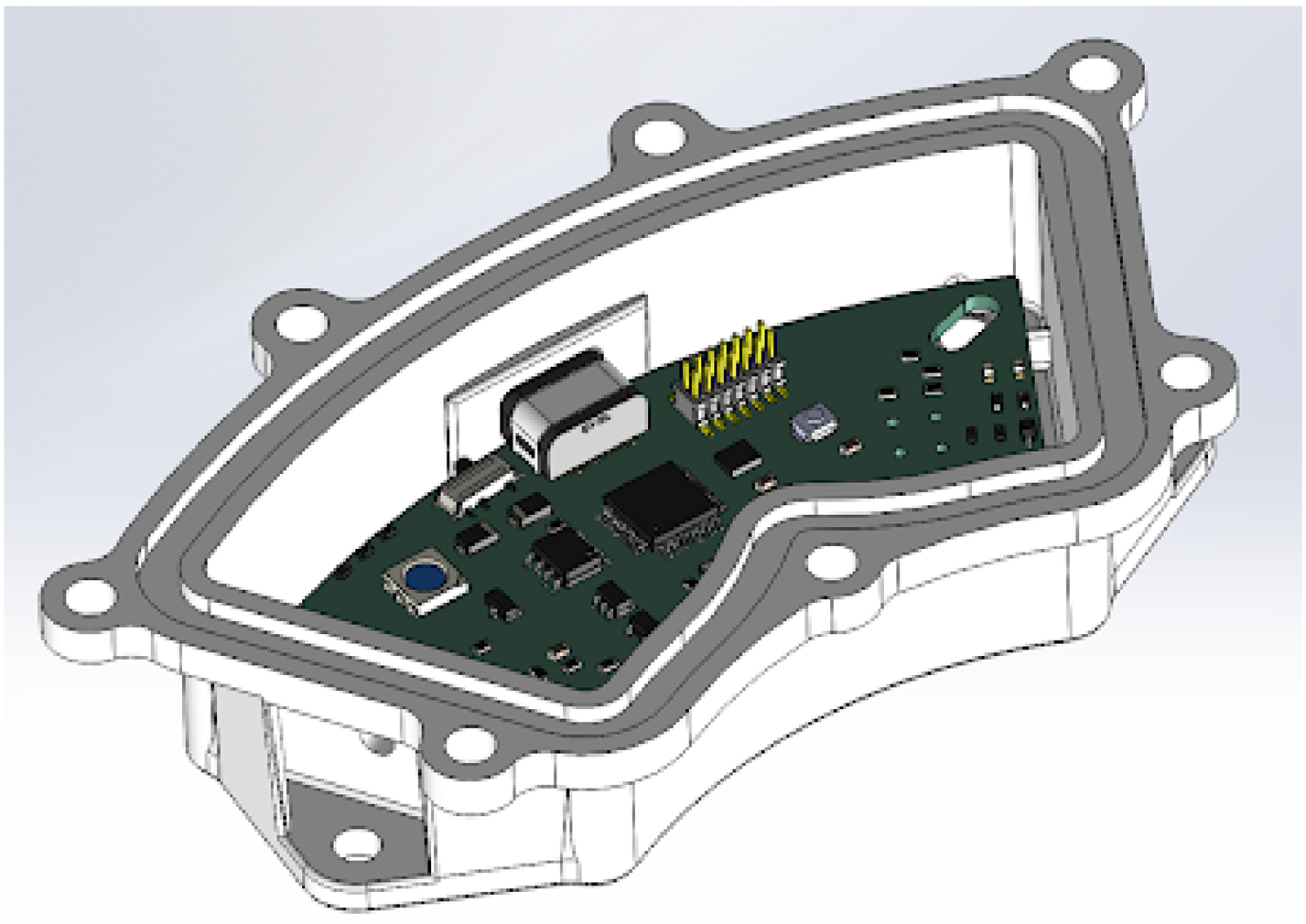
Installed rear-wheel package



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Sealed PCB enclosure CAD



Magnet-carrying preload nut



Magnetic Flux Requirment Calculation Sheet

L	W	H	V
0.125	0.25	0.25	0.0078125
			1.28E-07

Description	Variable	Value
Inputs		
Rated Magnetic Flux Density (Gauss)	B_r	12400
Volume of Magnet (m^3)	V	1.28E-07
Magnetic Flux density sensor requirem	β	5
distance from magnet centroid to sens	z	14.7

Outputs		
Magnetic Flux density sensor requirement (G)	β	50
distance from magnet centroid (m)	z	0.0147
Rated magnetic strength product	σ_r	1.59E-03
Magnetic strength product (G*m^3)	σ	0.000998
required distance with given magnet (mm)	z	**17.16**

NASA MODULAR LUNAR FLAGPOLE

NASA Johnson Space Center, ER4 | Jul–Aug 2025

Project Scope: Owned the deployment mechanism, material selection, detailed CAD, prototyping, and physical manufacturing for a modular lunar flagpole designed for tool-free operation by a single astronaut.

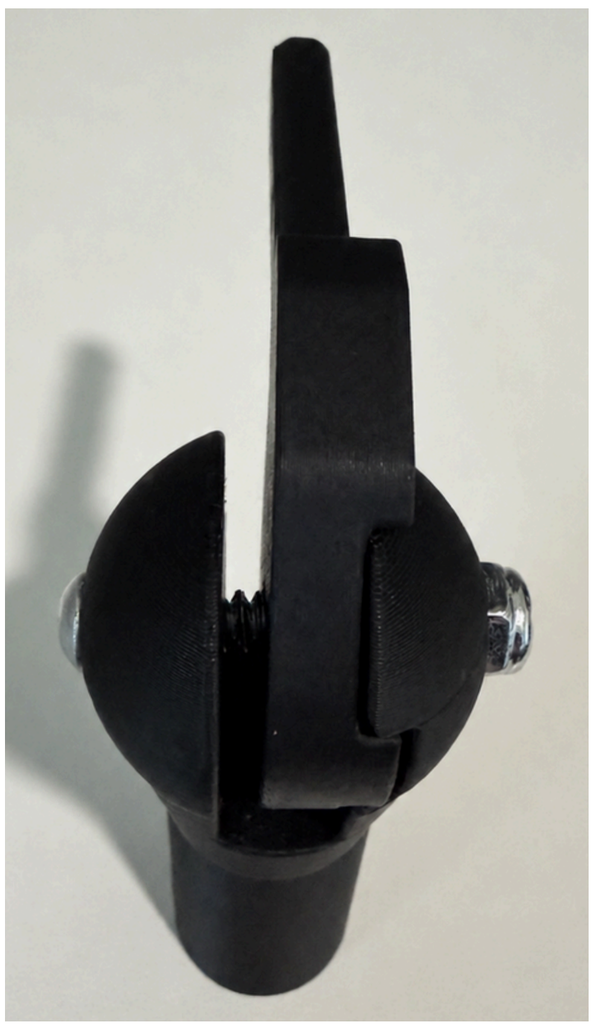
- Spring-loaded clevis pivot joint: single-motion deployment, auto-locks , no tools
- Chose 6061 aluminum over carbon fiber: avoids thermal degradation, outgassing, and dust-abrasion risk in vacuum; better machinability and cost
- Governing load case is astronaut handling/assembly force, not wind (no atmosphere present); joint sized for that despite flag self-weight reduced under 1/6g lunar gravity
- Delivered final prototype to NASA JSC ER4 robotics team

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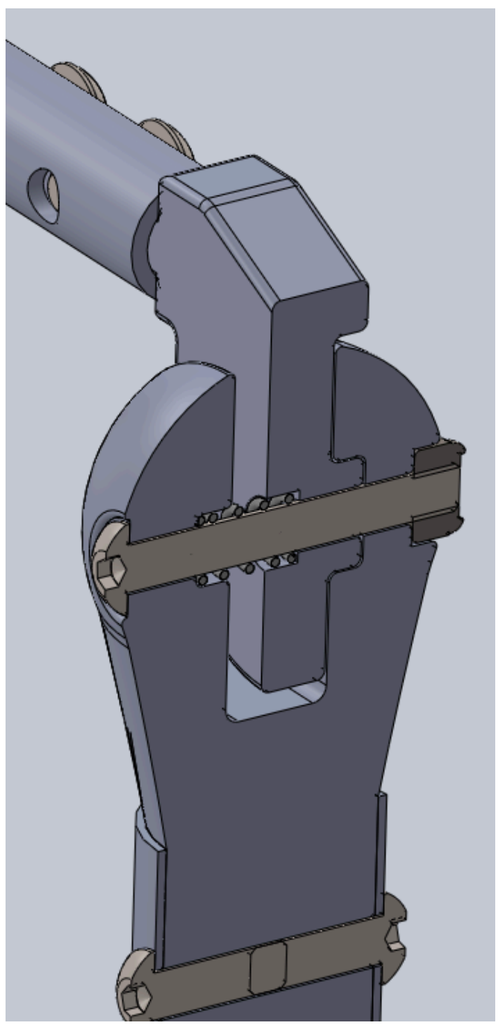
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Constraint	Result
Weight	<3 lb
Deployment	60 in, single motion, tool-free
Material	6061 aluminum (vs. carbon fiber)
Validation	Repeated manual deploy/handling cycles, 0 disengagement, 0 permanent deformation

3D Printed Proto



Spring-lock mech



VEX COMPETITION ROBOT SYSTEM DEVELOPMENT

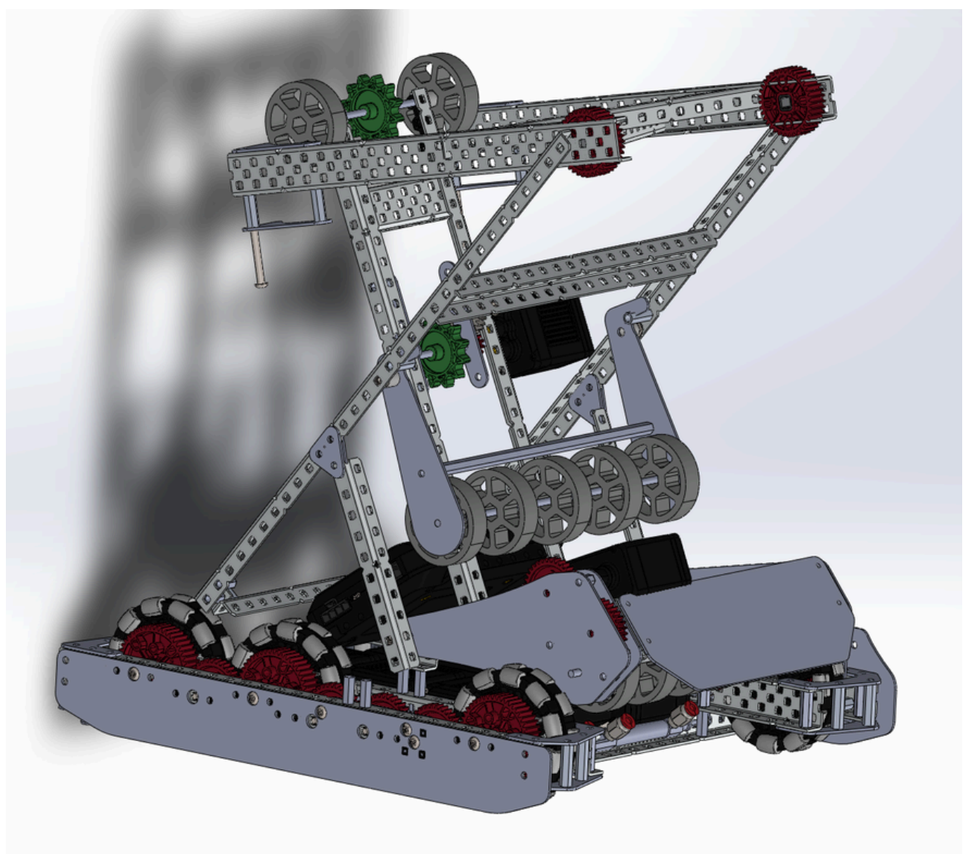
VEX Team 4148Z | Jan–Apr 2024

Project Scope: Owned the complete design and integration of a ring intake/transfer system and an indexed power take-off (PTO) endgame climb mechanism, including CAD, machining, assembly, programming, testing, and competition iteration.

- CNC-machined polycarbonate componets load-driven material removal for mass + deflection cut
- Piston-actuated sliding-gear PTO: low backlash drive↔climb shift, no added motor count
- 85% of robot mass packaged in bottom 2 in, improves acceleration, cornering stability, tip resistance

Result	Value
Intake mass	–15%+
Tower Deflection	0.10 → 0.06 in
Subsystem swap time	under 3 min
Shaft/bearing tolerance	±0.002–0.004 in
Chassis mass (bottom 2 in)	>85% of total robot mass

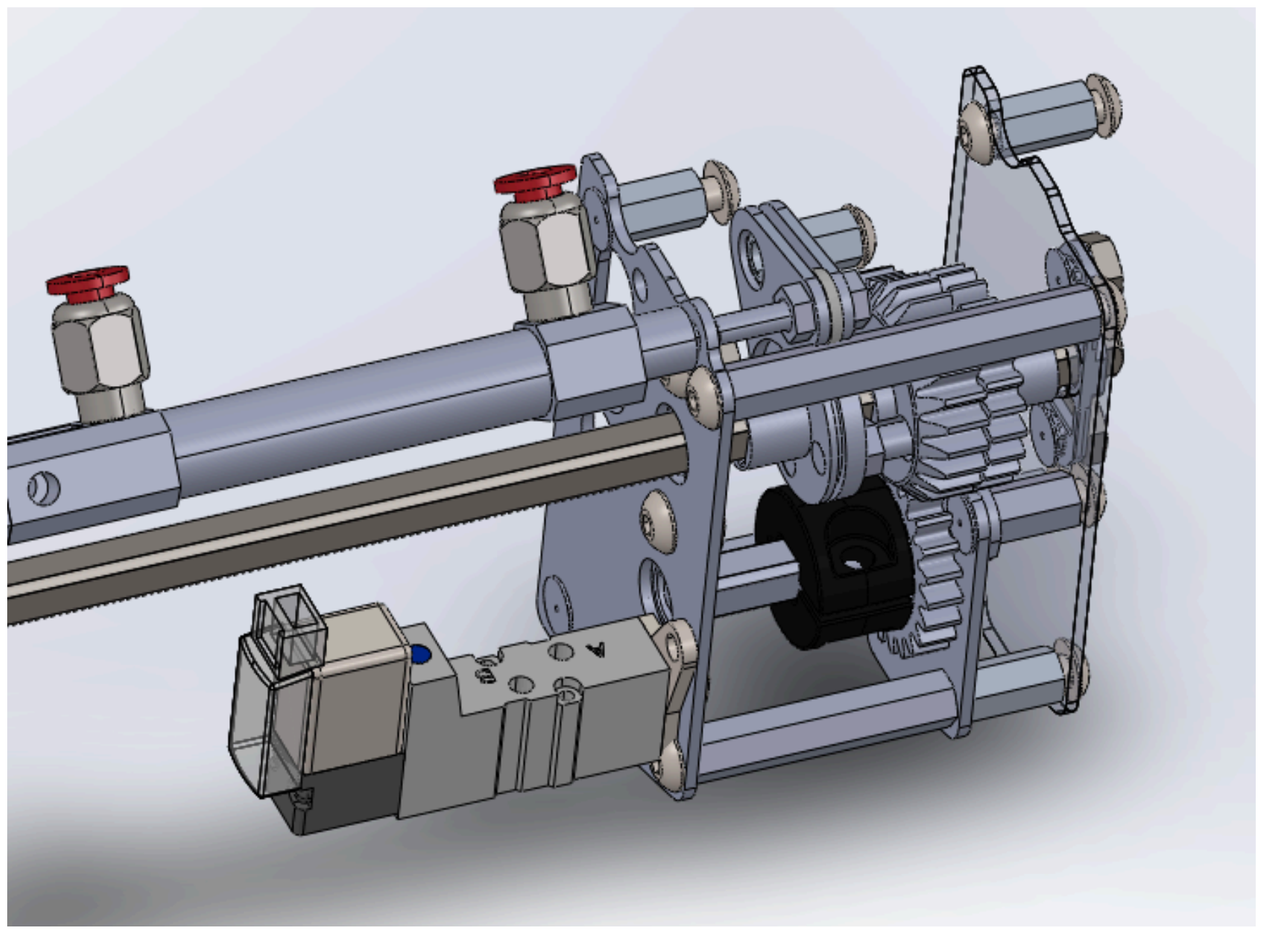
Robot CAD Assembly



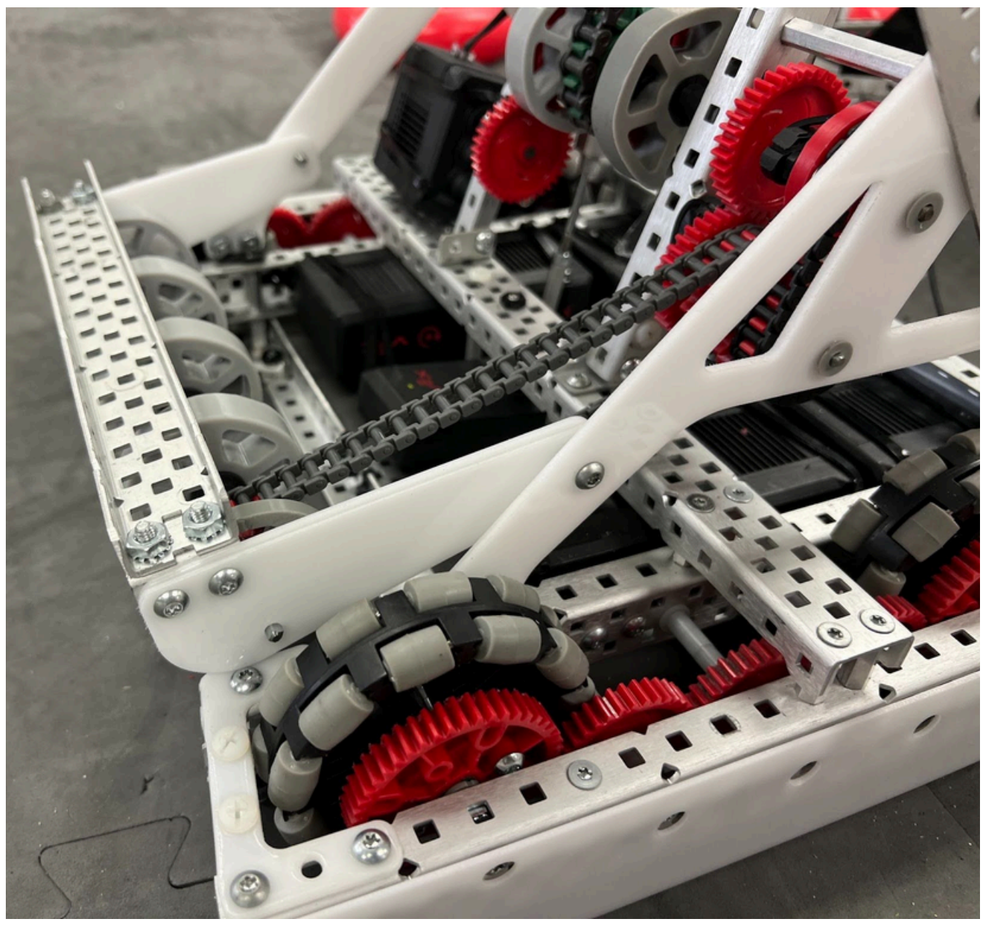
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PTO CAD Assembly (Half)



Physical Robot Assembly

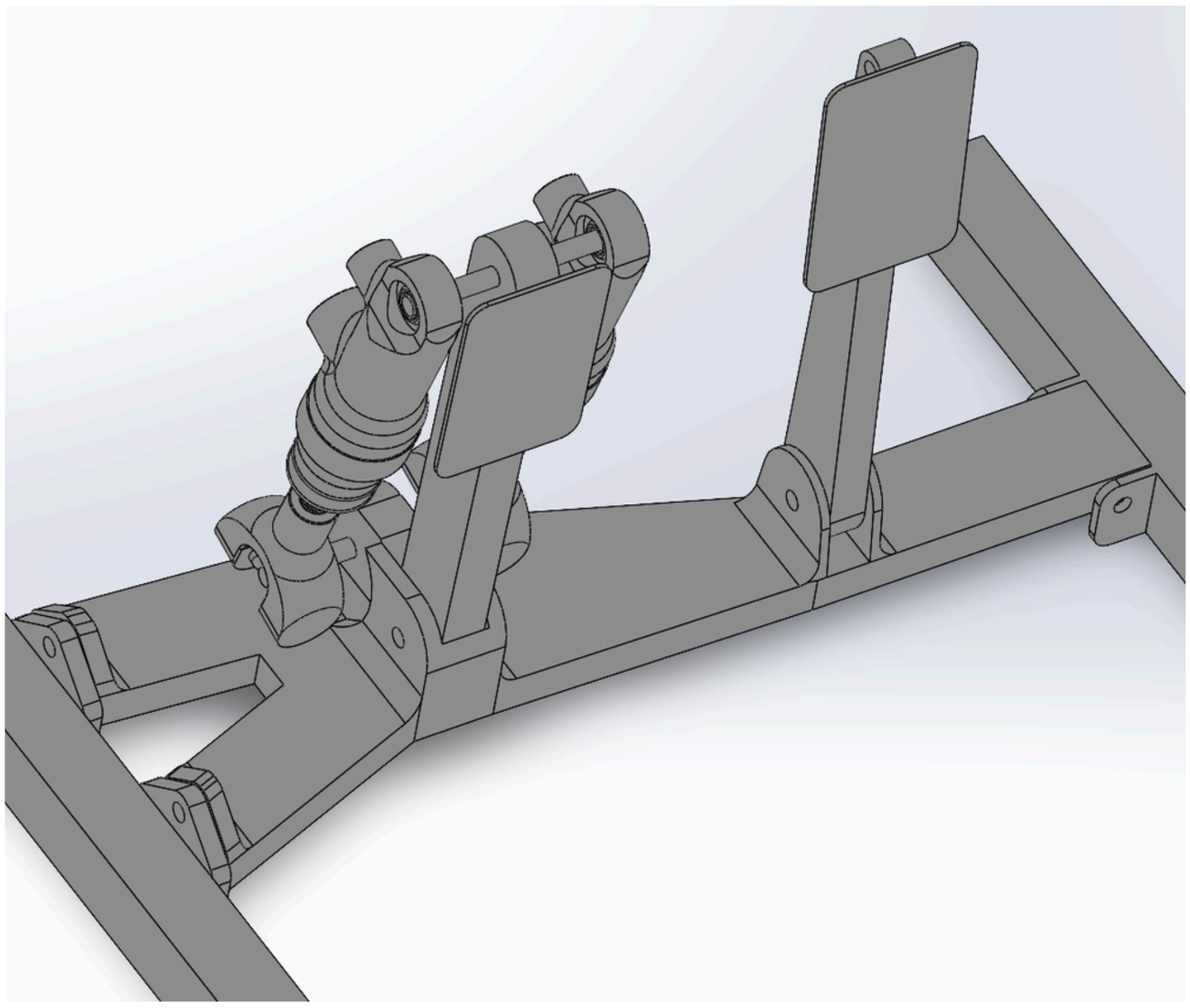


FSAE PEDAL BOX DEVELOPMENT

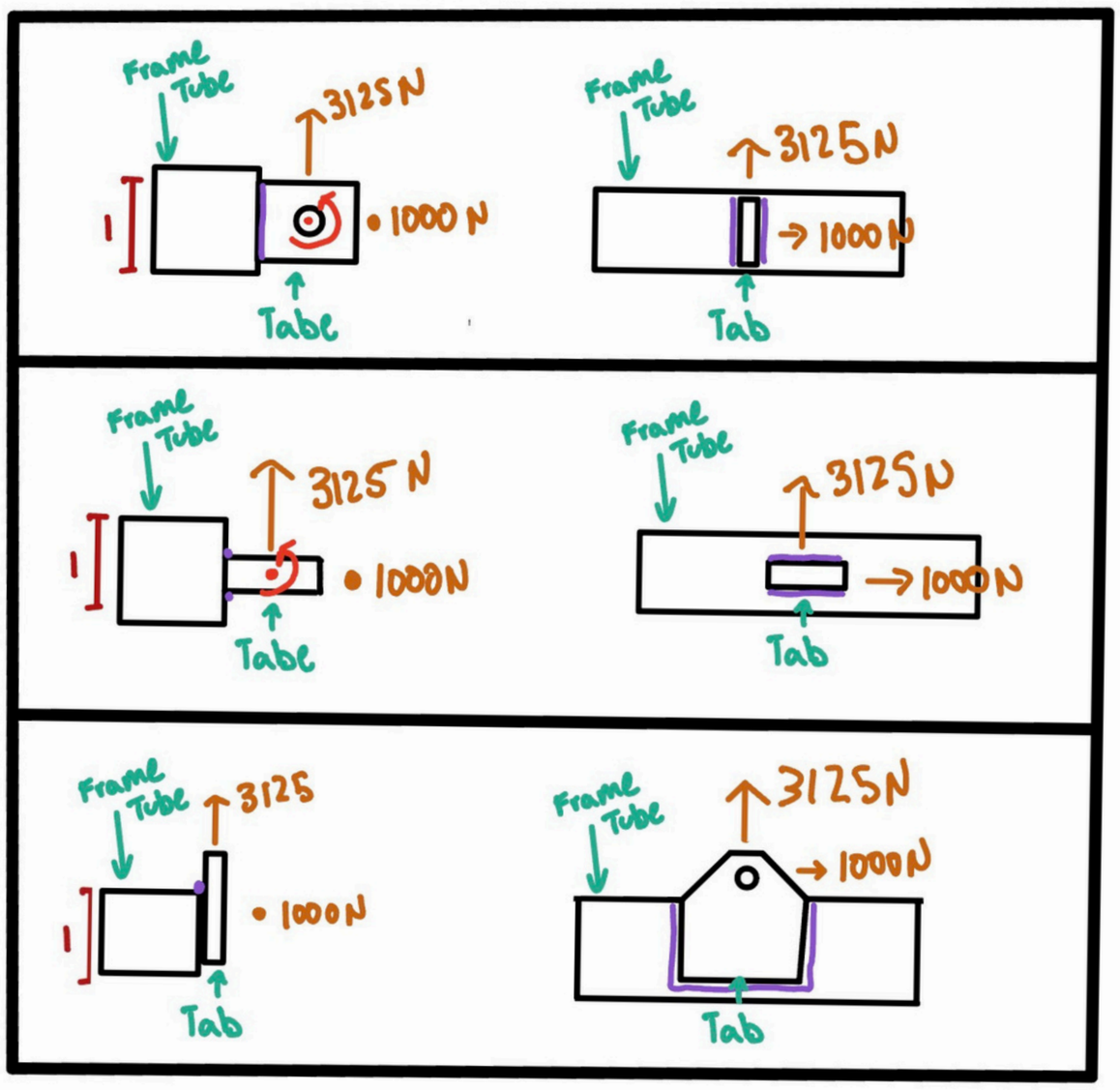
Longhorn Racing Electric | Aug 2026–Present

Project Scope: Own the requirements, architecture, and design of the team's pedal box, including hydraulic sizing, driver packaging, structural/load-path analysis, manufacturing, and validation.

- SolidWorks + Excel design study to solve pedal ratio, drove the vertical MC call
- Vertical MC adds ~0.75 in of length vs. horizontal's ~2 in last year, frees up front-bulkhead space
- Shrunk MC moment arm 11 → 6.25 in, cutting load and weight through the system
- FBD-driven tab study: reverse-engineered last year's loads, sized this year's tab and weld
- Excel brake-sizing model feeds directly into the team's live brake calc sheet



Initaial Space Claim Cad

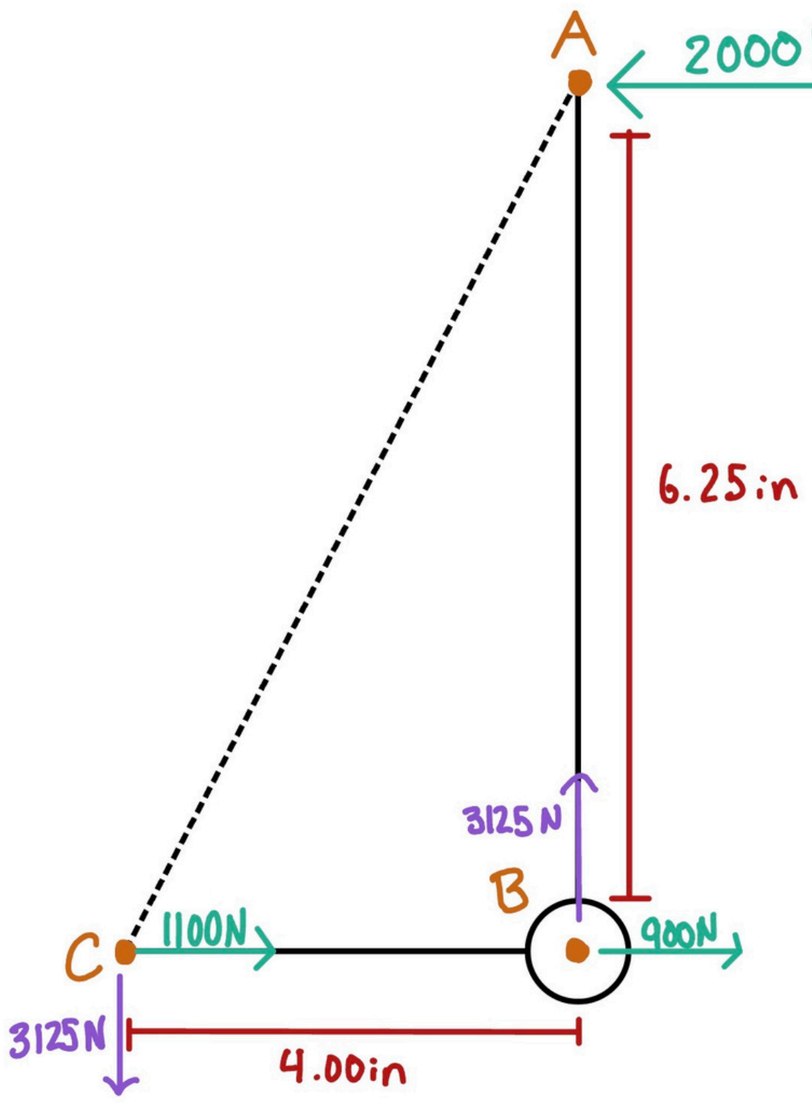


Tab Orentation FBD Comparison

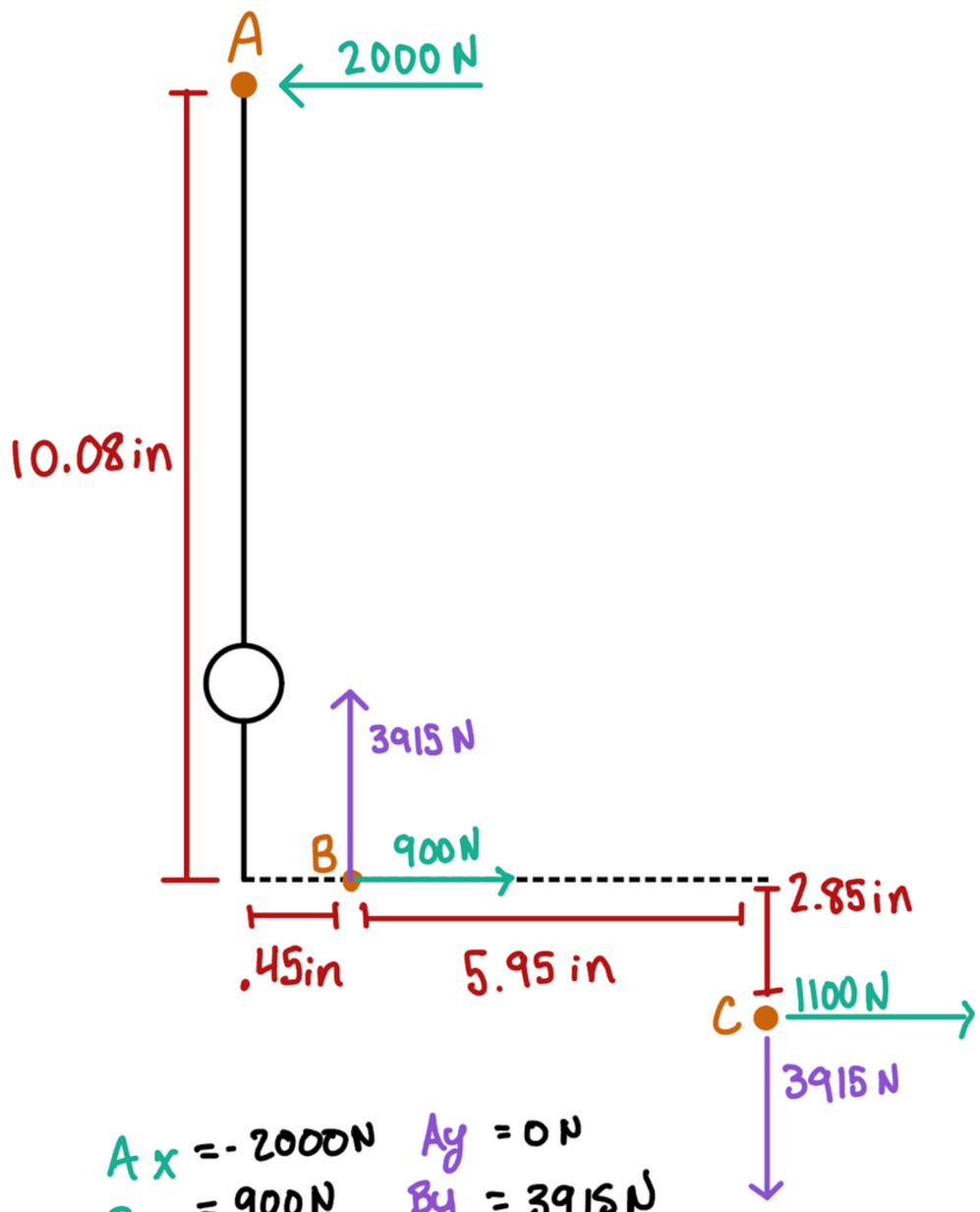
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Proposed Geometry Vertical Vs Horizontal FBD (Side Veiw)



Ax =-2000N Ay =0N
Bx =900N By =-3125N
Cx =1100N Cy =3125N



Ax =-2000N Ay =0N
Bx =900N By =3915N
Cx =1100N Cy =-3915N